



< Previous

✓

Next >

5.1.2 Overview

🔖 Bookmark this page

Pursue a verified certificate

✕

Verified Certificate

THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

now

Maurilio Pugliesi

Professor

UniversityX

now

Justine Doe, Ph.D.

Professor

UniversityX

edX

Verified Certificate

Valid Certificate ID

Issued Month / Year

12345678901234567890

→ Unlimited access to all course materials

→ Shareable [verified certificate](#)

→ AI learning assistant and translations

Upgrade for \$99

- **5.1 Opening**
 - 5.1.1 Of Gaussian elimination and LU factorization
 - 5.1.2 Overview
 - 5.1.3 What you will learn
- **5.2 From Gaussian elimination to LU factorization**
 - 5.2.1 Gaussian elimination
 - 5.2.2 LU factorization: The right-looking algorithm
 - 5.2.3 Existence of the LU factorization
 - 5.2.4 Gaussian elimination via Gauss transforms
- **5.3 LU factorization with (row) pivoting**
 - 5.3.1 Gaussian elimination with row exchanges
 - 5.3.2 Permutation matrices
 - 5.3.3 LU factorization with partial pivoting
 - 5.3.4 Solving $Ax = y$ via LU factorization with pivoting
 - 5.3.5 Solving with a triangular matrix
 - 5.3.6 LU factorization with complete pivoting
 - 5.3.7 Improving accuracy via iterative refinement
- **5.4 Cholesky factorization**
 - 5.4.1 Hermitian Positive Definite matrices
 - 5.4.2 The Cholesky Factorization Theorem
 - 5.4.3 Cholesky factorization algorithm (right-looking variant)
 - 5.4.4 Proof of the Cholesky Factorization Theorem
 - 5.4.5 Cholesky factorization and solving LLS
 - 5.4.6 Implementation with the classical BLAS
- **5.5 Enrichments**
 - 5.5.1 Other LU factorization algorithms
- **5.6 Wrap Up**
 - 5.6.1 Additional homework
 - 5.6.2 Summary

Discussion

Topic: Week 05 / 05.1.2

Hide Discussion

Add a Post

Add a Post

Show all posts

by recent activity

There are no posts in this topic yet.

< Previous

Next >



edX

- About
- Affiliates
- edX for Business
- Open edX
- Careers
- News

Legal

- Terms of Service & Honor Code
- Privacy Policy
- Accessibility Policy
- Trademark Policy
- Sitemap
- Cookie Policy
- Your Privacy Choices

Connect

- Idea Hub
- Contact Us
- Help Center
- Security
- Media Kit

